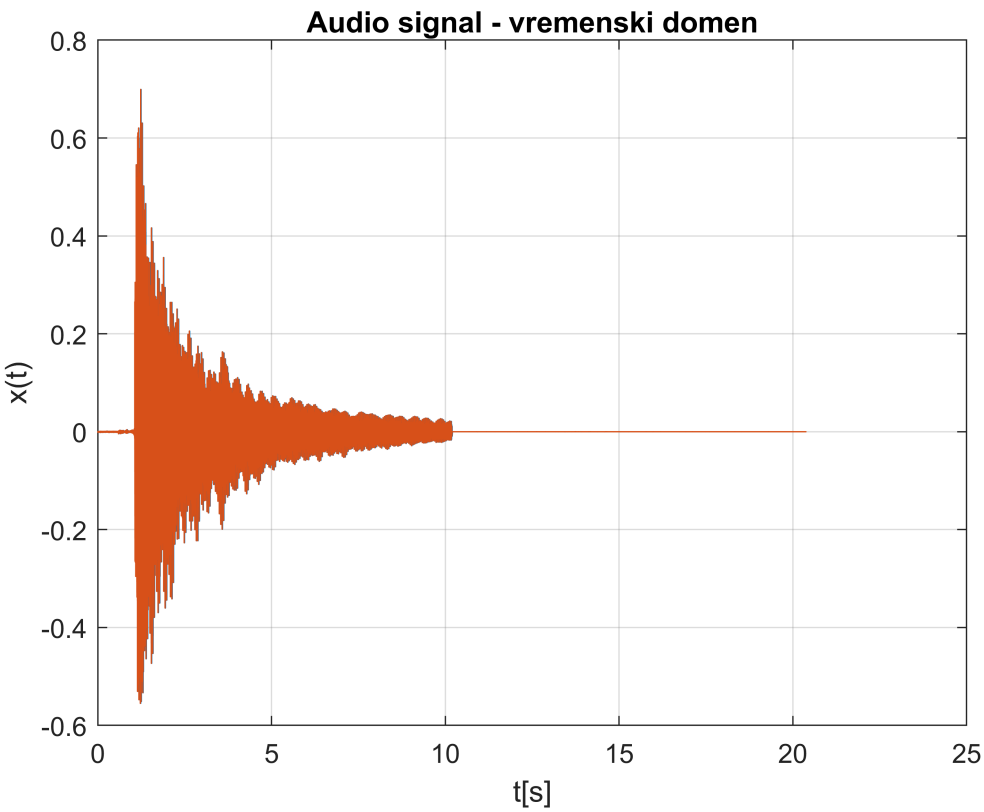
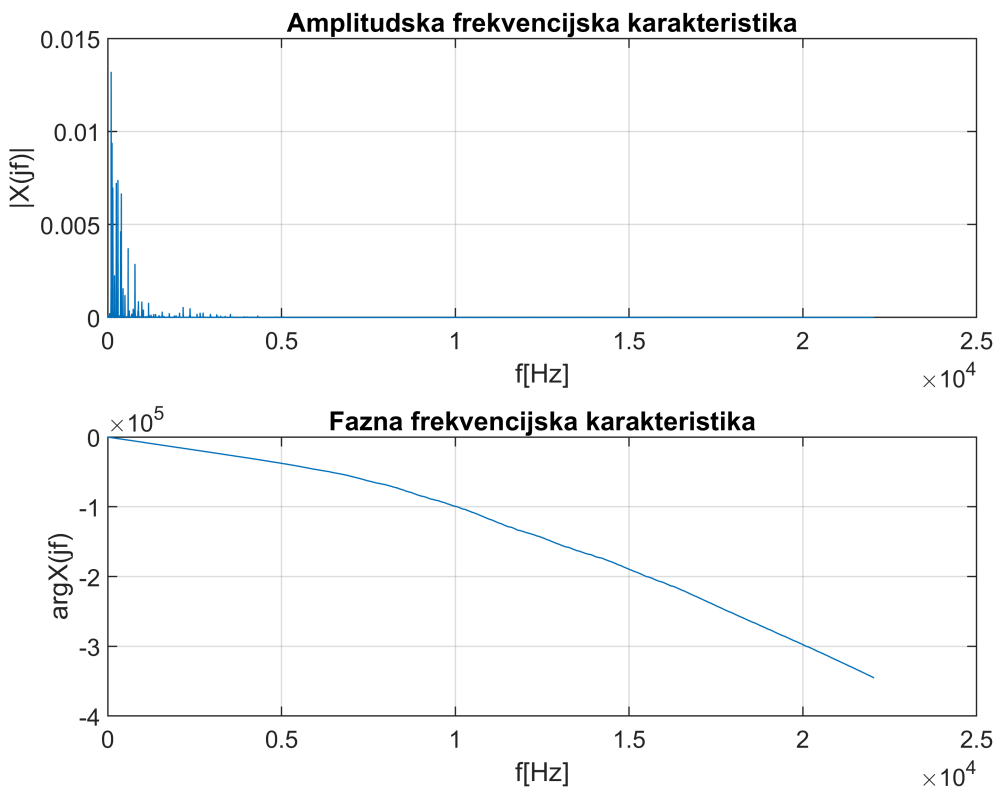
```
clear all; close all; clc;

%akord Q = 0

filename = 'Akord - Q0.wav'; %ucitavanje signala iz fajla
[x, Fs] = audioread(filename);
%-------------------------------------------
t = 0 : 1/Fs : (length(x) - 1)/Fs;

figure(1);
plot(t, x);
xlabel('t[s]'); ylabel('x(t)'); title('Audio signal - vremenski domen'); grid on;
```




```
%-------------------------------------------
N = 2^nextpow2(length(x));
X = fft(x, N)/length(x);

f1 = 0 : Fs/N : Fs/2;
X1 = abs(X(1 : N/2+1)); %amplituda
X1(2 : N/2+1) = 2 * X1(2 : N/2+1);
Xphase = unwrap(angle(X(1 : N/2+1))); %faza

figure(2);

subplot(2, 1, 1);
plot(f1, X1);
```

```
xlabel('f[Hz]'); ylabel('|X(jf)|');
title('Amplitudska frekvencijska karakteristika'); grid on;

subplot(2, 1, 2);
plot(f1, Xphase);
xlabel('f[Hz]'); ylabel('arg{X(jf)}');
title('Fazna frekvencijska karakteristika'); grid on;
```




```
%---------------------------------------------
[maxPeaks, frequency] = findpeaks(X1, f1, 'NPeaks', 3, 'SortStr', 'descend');

% Najveca 3 pika su nam: maxPeaks = [0.0132,0.0094,0.0074],
% sa frekvencijama: frequency = [98.1611 Hz, 123.6477 Hz, 293.0955 Hz]

% 98.1611 => 98 => G2
% 123.6477 => 123 => B2
% 293.0955 => 293 => D4
%---------------------------------------------
% Kako smo dobili note G, B, D, zakljucujemo da se radi o G akordu
```